

FREEDOM INTERNATIONAL SCHOOL
WORKSHEET
PHYSICS
OSCILLATIONS
CLASS XI

1. What will the period of oscillation be, if the length of a second's pendulum is halved?
2. A spring having a force constant k is divided into three equal parts. What would the force constant be for each individual part?
3. The maximum velocity of a particle, executing simple harmonic motion with an amplitude of 7 mm, is 4.4 m/s. What is the period of oscillation?
4. Two simple harmonic motions are represented by the equations:
 $x_1 = 5 \sin(2\pi t + \pi/4)$ and $x_2 = 5\sqrt{2} (\sin 2\pi t + \cos 2\pi t)$. What is the ratio of their amplitudes?
5. A point particle of mass 0.1 kg is executing SHM of amplitude 0.1 m. When the particle passes through the mean position, its kinetic energy is 8×10^{-3} J. Obtain the equation of motion of the particle if the initial phase of oscillation is 45° .
6. A spring of force constant 1200 N/m is mounted horizontally on a horizontal table. A mass of 3.0 kg attached to the free end of the spring, is pulled sideways to a distance of 2.0 cm and released. (i) What is the frequency of oscillation of the mass? (ii) What is the maximum acceleration of the mass? (iii) What is the maximum speed of the mass?
7. Two bodies M and N of equal masses are suspended from two separate massless springs of spring constants K_1 and K_2 respectively. If the two bodies oscillate vertically, such that their maximum velocities are equal, then, find the ratio of the amplitude of M to that of N.
8. Two simple harmonic motions are represented by the equations:
 $y_1 = 0.1 \sin(100\pi t + \pi/3)$ and $y_2 = 0.1 \cos \pi t$
What is the phase difference of the velocity of the particle 1 with respect to the velocity of particle 2?
9. If the earth were a homogeneous sphere of radius R and a straight hole bored in it through its centre, show that a body dropped into the hole will execute SHM and find its time period.
10. One end of a U-tube containing mercury is connected to a suction pump and the other end is connected to the atmosphere. A small pressure difference is maintained between the two columns. Show that when the suction pump is removed, the liquid in the U-tube executes SHM.